

## 33 — LNG carrier: L3 variation 1 000 kW, completing the lookup table

**Proves** · An LNG vessel picking up its 1 000 kW Level 3 variation from the lookup.

### Mechanics this scenario turns on

- When the Level 3 variation field is empty the backend looks it up from the vessel type (Bulk 250 · Container 1 500 · LNG 1 000 · otherwise the 500 default), all from `appsettings.json`.
- Level 3 (DRC) damps the hotel/mission swing:  $\text{variation} \times 0.8$ , minus whatever the battery already shaved, so the same kilowatt is never counted twice.
- ME and AE each use **their own fuel's** CO2 factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753 · Ammonia 0.35154). The two per-engine cards must sum to the panel total.

**Panels described below** · Verified figures · The linearity check this enables

**Anything not described here** — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

**Trust** · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

**Read after** · scenario 14.

The last of the three `VesselVariations` entries. With 14 (Bulk Carrier 250), 25 (Container 1 500) and 26 (default 500), the table and its fallback are now fully covered.

## Verified figures

|                           |            |
|---------------------------|------------|
| variation                 | 1 000.0 kW |
| reduced ( $\times 0.80$ ) | 800.0 kW   |
| L3 savings                | 92.83 t/yr |

Same plant as 19/25/26, so the only moving part is the vessel name.

## The linearity check this enables

Three points on one plant, from three different lookup routes:

|                       |                                  |
|-----------------------|----------------------------------|
| 500 kW → 45.25 t/yr   | (ratio 0.0905 t per kW of swing) |
| 1000 kW → 92.83 t/yr  | ( 0.0928)                        |
| 1500 kW → 142.79 t/yr | ( 0.0952)                        |

Savings rise slightly **faster** than the swing. That is the expected shape: DRC earns its money from the curvature of the SFOC curve, and a wider swing pushes the down-stroke further into the steep low-load region where the curve bends hardest.

If a future change made these three points exactly proportional, the curvature term would have been lost somewhere — the three scenarios together detect that, where any one alone would not.

**Takeaway:** covering a lookup table means one scenario per entry *and* checking they relate to each other sensibly. The relationship is the stronger assertion.